

DAY – 8 QUESTION

Basic Test Case Write a unittest class TestMath with one test method that checks if 2 + 3 equals 5.
Setup and Teardown Create a test class that uses setUp() to initialize a list [1, 2, 3] and tearDown() to print "Test completed". Verify that the list length is 3 inside your test method.
Multiple Assertions Write a test class TestStringMethods with methods to test: "hello".upper() equals "HELLO" "hello".isupper() returns False
Exception Testing Use assertRaises to verify that dividing by zero (10 / 0) raises a ZeroDivisionError.
Test Suite Execution Create two test classes (TestAdd and TestSubtract) and combine them into a single test suite using unittest.TestSuite. Run the suite using unittest.TextTestRunner().

CODE

```
import unittest
class TestMath(unittest.TestCase):
    def test_add(self):
        self.assertEqual(2 + 3, 5)
class TestList(unittest.TestCase):
    def setUp(self):
        self.sample_list = [1, 2, 3]
    def tearDown(self):
        print("Test completed")
    def test_list_length(self):
        self.assertEqual(len(self.sample_list), 3)
class TestStringMethods(unittest.TestCase):
    def test_upper(self):
        self.assertEqual("hello".upper(), "HELLO")
    def test_isupper(self):
        self.assertFalse("hello".isupper())
class TestExceptions(unittest.TestCase):
    def test_divide_by_zero(self):
        with self.assertRaises(ZeroDivisionError):
            result = 10 / 0
class TestAdd(unittest.TestCase):
    def test_addition(self):
        self.assertEqual(5 + 5, 10)
class TestSubtract(unittest.TestCase):
    def test_subtraction(self):
        self.assertEqual(10 - 5, 5)
if __name__ == "__main__":
    suite = unittest.TestSuite()
    suite.addTest(unittest.makeSuite(TestAdd))
    suite.addTest(unittest.makeSuite(TestSubtract))
    runner = unittest.TextTestRunner()
    runner.run(suite)
```

OUTPUT

```
===== test session starts =====
collecting ... collected 7 items

unittest_assign.py::TestMath::test_add PASSED [ 14%]
unittest_assign.py::TestList::test_list_length PASSED [ 28%]Test completed

unittest_assign.py::TestStringMethods::test_isupper PASSED [ 42%]
```

```
unittest_assign.py::TestStringMethods::test_upper PASSED [ 57%]  
unittest_assign.py::TestExceptions::test_divide_by_zero PASSED [ 71%]  
unittest_assign.py::TestAdd::test_addition PASSED [ 85%]  
unittest_assign.py::TestSubtract::test_subtraction PASSED [100%]
```

```
===== 7 passed in 0.04s =====
```

Process finished with exit code 0